SEAGAS NLP - Remaining Work and Known Issues

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COS40005 Computing Technology Project A | POC004 | Code: dev at dd91def

1. Where the NLP Stands

The NLP now works inside the system, but it is not finished against the brief. It has moved from "the system cannot use it" to "the system uses it". What is left is proving it is accurate (evaluation) and making it more accurate (fixing known errors), which is where the marks are.

Done	Status
NLP module (Yong Hin)	Keyword, semantic and hybrid extraction, and the Strong / Developing / Gap matcher; 321 tests pass.
Connected to the system	Since 16 October (commit 09eaad9 on dev): pasted JDs are analysed, assessments use the matcher, every skill shows its reason.
Semantic packages	Installed and verified on Carl's machine; other machines fall back to keyword until `nlp/requirements.txt` is installed.
First comparison	Three methods on 65 real JDs: keyword 9.2, semantic 7.3, hybrid 11.2 skills per JD.

2. What Is Still Missing

#	Item	Owner	Priority
1	Accuracy evaluation: gold labels and precision / recall / F1	Yong Hin + 1 member	P0
2	Known misjudgments: generic words and missing aliases	Yong Hin	P1
3	Rule proposals A / B / C	Whole team	P1
4	JD dataset too small (65 of 180 target)	Yong Hin	P1
5	TF-IDF baseline (fourth comparison method)	Yong Hin	P2
6	Evidence rule has no input on the website	Ren Hang / Shanice (+ Carl)	P1
7	CV parsing (optional)	Yong Hin	Optional

3. Details

3.1 Accuracy evaluation (most important)

Aspect	Detail
Problem	The brief's core question is which matching method is most accurate. That needs hand-labelled gold data to compute precision, recall and F1. There are **0** gold labels: `gold_labels.csv` has only a header.
Why it matters	Without gold labels we can only say the methods give different results (keyword and semantic agree with a mean Jaccard of 0.51), not which one is right. Deliverable 11 (AI evaluation results) depends on this.
Next step	Label 20-30 of the 60 sampled JDs first; a second member labels 12 of them to compute Cohen's kappa. The comparison script already computes precision / recall / F1 once labels exist.

3.2 Known misjudgments

Aspect	Detail
Generic words	Generic words are matched to specific tools: "testing" to A/B Testing (14 JDs), also "analytics", "database" and "programming languages".
Missing aliases	No alias for "UI/UX", "firewall(s)"; "web-development" does not match React, JavaScript or HTML.
Too broad	Aliases such as "Initiative" and "Monitoring" match unrelated text ("initiatives" to Self-Motivation in 9 JDs).
Word forms	"meet deadlines" in a JD does not match the alias "Meeting deadlines".
Real example	A Software Developer internship JD (1,831 characters) tested on the website on 16 October: keyword found only Business Analysis, Problem Solving, Git and Documentation. Hybrid added A/B Testing (from "testing") and Log Analysis (from "change logs"), both wrong. A student who lists A/B Testing would wrongly get Strong. "Basic knowledge of programming, databases and web-development concepts" matched nothing.
Next step	Filter one-word generic candidates in semantic extraction, add the missing aliases, tighten the broad ones, handle verb forms; then re-run the 65-JD comparison.

3.3 Rule proposals A / B / C (team decision)

Aspect	Detail
A	Two different skills that are similar count at most as Developing. Today Teamwork from Leadership (similarity 0.600) counts as Strong (104 rows in the before / after comparison).
B	Tools of the same kind count as Developing. Tableau scores only 0.251 against Power BI, and 305 rows of related skills (e.g. SQL and PostgreSQL) became Gap.
C	A beginner skill with only coursework evidence counts as Developing (1,046 of 6,789 rows).
Next step	Decide at the next team meeting; Yong Hin changes `nlp/matcher.py` and the rules document. No backend change is needed.

3.4 JD dataset too small

Aspect	Detail
Problem	65 JDs (10-13 per role). The brief suggests 100-500 job adverts; the Sprint 3 target is 30 per role (180 in total).
Next step	Collect more JDs (public APIs or job sites), re-run the comparison, and adjust the role skill templates from the skill-frequency analysis.

3.5 TF-IDF baseline

Aspect	Detail
Problem	The task breakdown and the brief list TF-IDF + cosine similarity, but it is not implemented.
Next step	About half a day with scikit-learn (already in nlp/requirements.txt). It extends the comparison to keyword / TF-IDF / semantic / hybrid.

3.6 Evidence rule cannot take effect

Aspect	Detail
Problem	The matcher uses evidence (project, certification, coursework, internship): a beginner skill with no evidence is Developing; intermediate or advanced, or any evidence, is Strong. The website has no field for evidence when adding a skill, so for real users beginner is always Developing and intermediate or above is Strong just by self-rating.
Status	The backend already accepts `evidence_type` on `POST /students/skills` (validated since 16 October). Only the frontend is missing.
Next step	Add an evidence choice in My Profile (Ren Hang / Shanice). Better still, derive evidence from projects (run `extract_skills()` on their technologies and description) and certificates; this also answers the brief's question of whether projects and certificates improve employability.

3.7 CV parsing (optional)

Aspect	Detail
Problem	Not started.
Next step	Reuse `extract_skills()` on uploaded CV text to suggest skills for the profile. If an LLM is used for explanations (Level 3), set privacy rules first: no student data to external APIs, or remove identifying information.

4. Suggested Order

When	Task
Before 24 Oct	Put the 65-JD comparison and the rules / thresholds documents into the Sprint 2 report as evidence; note honestly that gold-label evaluation is planned for Sprint 3-4.
Sprint 3	Fix generic words and aliases (3.2); decide rules A / B / C (3.3); evidence input (3.6); start gold labels (3.1); grow the JD set (3.4).
Sprint 3-4	Finish gold labels and kappa; add TF-IDF (3.5); compute precision / recall / F1 for all four methods; optional CV parsing (3.7).